TECHNICAL EVIDENCE REPORT

Automated detection and visualisation of Tn1, Tn2 and Tn3

Expert rules, real-accession validation, locus maps, locus tables and hierarchical outputs

Definition set	Software status	Report date	Validation
2026-08-27	External alpha	27 August 2026	14/14 checks pass

Purpose

This document is the reviewable evidence package: it explains what is run, how the biological rules are specified, and what the program produced from the reviewed accession sequences. The actual figures and tables are embedded in the report.

Scope statement

The validated scope is Tn1, Tn2, Tn3 and the reviewed close definitions listed here. It is not yet an exhaustive replacement for expert mobile-element annotation across every family.

Contents and claims at a glance

What this report demonstrates

- An arbitrary multi-FASTA input can be scanned without pre-existing annotations.
- Tn1/Tn2/Tn3 components are detected independently and assembled using an explicit component grammar.
- A complete locus is classified from its component grammar and weighted local component profiles; a whole-element comparison is secondary confirmation.
- Every result can be inspected as a locus map, a locus table, the original hierarchy view, JSON, GFF3 and CellGen format.
- The seven reviewed accession definitions produce complete component grammars and exact declared calls; pEK499 produces two retained partial Tn2-like fragments rather than an invented complete Tn2.

Important interpretation

The seven exact accession runs are positive controls against the sequences used to define those names. They prove correct rule execution, component recovery and output generation. They do not by themselves estimate sensitivity on a broad independent population. The natural pEK499 fragments, synthetic variation tests and related-element negative controls test behaviour away from exact references.

1. What the tool does

Matryoshka is a per-sequence nested mobile-element annotator. For this report, its validated task is to find Tn1/Tn2/Tn3-group components in a FASTA sequence, assemble candidate loci, assign a reviewed name only when the evidence satisfies the expert rules, and render the same evidence in several complementary formats.

Input to output

Stage	Operation	Why it matters
1	Read FASTA	One or many contigs or complete sequences; no feature annotation is required.
2	Detect components	Scan independently for 38 bp terminal inverted repeats, blaTEM, tnpR, res and tnpA.
3	Assemble loci	Group nearby features and test the required order and orientation on either strand.
4	Classify by expert rules	Score the detected component profiles and assign a type only when the declared threshold and margin are met.
5	Compare references	Optionally confirm an exact reviewed definition and record secondary closest-reference context.
6	Write outputs	JSON, GFF3, CellGen, hierarchy SVG, locus map, locus table and proof ledger.

Call classes

Visible result	Meaning
Tn1 / Tn2 / Tn3 or named subtype	Complete component grammar and exact match to a reviewed declared definition.
Tn1-like / Tn2-like / Tn3-like	Complete grammar and a clear best weighted component profile, without requiring a complete-element match.
Tn1/2/3 fragment	At least 400 aligned reference bases, but an end or required component is missing or the grammar is incomplete.
Unresolved Tn1/Tn2/Tn3-group unit	The group grammar is supported but the component-profile score or type margin is insufficient.
No Tn1/Tn2/Tn3 call	Some shared components may still be annotated and drawn, but the element-level rule is not satisfied.

2. What is actually run

Built-in analysis

The command scans independently for the component sequences and assembles their collinear matches across substitutions, insertions and deletions. The expert-rule classifier uses those components to establish the family and type. Under the default validated profile, a separate whole-element comparison is then attached as confirmation or context. The tn123-components profile omits that complete-element lookup entirely.

Optional detector layer

The --detectors option controls additional programs, not the built-in Tn1/Tn2/Tn3 scan. The report bundles were generated with --detectors none, which means ISEScan, AMRFinderPlus and IntegronFinder were not launched. Built-in BLAST detection and the expert-rule classifier still ran. With --detectors available or --detectors all, those programs add broader IS, AMR-gene and integron evidence.

Analysis	Role	Used for this report
BLAST component scan	Detect IRL/IRR, blaTEM, tnpR, res and tnpA independently	Yes; primary
Component assembler	Test count, order, orientation and ends	Yes
Expert-rule classifier	Assign family/type-like, fragment or unresolved result from weighted component profiles	Yes; primary
BLAST whole-locus comparison	Optionally confirm an exact reviewed definition or add closest-reference context	Yes; secondary
Boundary/direct-repeat check	Record evidence immediately outside inferred ends when flanks exist	Yes
ISEScan / AMRFinderPlus / IntegronFinder	Broader optional component discovery	Not launched

Reproducible commands

```
# Prove discovery without complete-element lookup
matryoshka run arbitrary-demo.fasta --out results/component-only \
  --profile tn123-components --detectors none

# Seven reviewed definitions with secondary exact confirmation
matryoshka run matryoshka/references/tn1_tn2_tn3.fasta \
  --out results/tn123-reviewed --detectors none

# Export the rules for expert review
matryoshka definitions --format markdown --out tn123-definitions-review.md
```

Reproducibility metadata

Each result directory includes run.json, the software and definition versions, reference profile, detector selection, input checksum, runtime parameters and a direct index of every locus map/table pair.

3. How the expert rules are specified

The biological definitions are data, not a collection of element-specific if-statements hidden in the Python source. The authoritative file is matryoshka/tn123_definitions.yaml. A JSON Schema checks its required structure, and the same content can be exported as YAML, JSON or a formatted Markdown review document.

Definition structure

Block	Human meaning
family	Shared biological description, expected terminal-IR length and target-site duplication length.
grammar	Required component counts and forward/reverse order.
component_detection	Identity, coverage, length and chaining thresholds for each component class.
classification	Weighted component-profile type rules, exact-reference confirmation and fragment rules.
types	Canonical Tn1, Tn2 and Tn3 sequences and their annotated component layouts.
definitions	Reviewable named sequences/subtypes, accession provenance, expert description and differences from parent.
related_element_policy	Explicit negative-control policy to prevent shared components being over-named.

Shared Tn1/Tn2/Tn3 grammar

```
terminal IR -> blaTEM -> tnpR -> res -> tnpA -> terminal IR
   38 bp       one       one     one    one      38 bp
```

The reverse-complement copy must contain the corresponding reversed order. This establishes group membership, but it does not distinguish Tn1, Tn2 and Tn3 because their structures and much of their sequence are shared.

Current thresholds

Decision	Rule
Exact declared definition	100% identity, 100% reference coverage, both ends, zero mismatches/insertions/deletions and complete component grammar.
Rule-based type	Complete grammar; weighted local component score at least 90%; best type exceeds the next by at least 0.5 points.
Component weights	blaTEM 1; tnpR 2; res 3; tnpA 3. The combined profile, rather than a single gene, assigns the type-like call.
Secondary reference	At least 95% identity and 80% coverage; used for context or exact confirmation, never to create the family/type.
Fragment	At least 400 aligned reference bases; retained without a complete name.

4. Worked expert definition: Tn1

Human-readable rule

Call a Tn1-like candidate when the six required components are independently detected in the correct order and their weighted local profiles best support Tn1 above the declared score and margin. Upgrade that call to exact reviewed Tn1 only when an optional secondary comparison is a complete, gap-free and mismatch-free match to NC_008 357. The canonical sequence carries blaTEM-2.

YAML representation used by the program


```
classification:
  type_assignment:
    method: weighted component profiles
    discriminator_role_weights:
      blaTEM: 1
      tnpR: 2
      res: 3
      tnpA: 3
    minimum_component_profile_score_percent: 90
    minimum_type_margin_percent: 0.5
  reference_comparison:
    role: secondary context after rule-based classification

types:
  Tn1:
    canonical_reference: Tn1_NC_008 357
    source_accession: NC_008 357
    components:
      - {role: terminal_IR, name: IRL, start: 1, end: 38}
      - {role: blaTEM, name: blaTEM-2, start: 148, end: 1008, strand: '-'}
      - {role: tnpR, name: tnpR, start: 1191, end: 1748, strand: '-'}
      - {role: res, name: res, start: 1754, end: 1867}
      - {role: tnpA, name: tnpA, start: 1911, end: −34, strand: '+'}
      - {role: terminal_IR, name: IRR, start: −38, end: −1}
```


Negative coordinates count from the right-hand end of the reference. This keeps the component layout readable and reusable for definitions of different lengths.

How to add or revise a subtype

- Add the public reference sequence and provenance.
- Add a definitions entry containing the parent type, visible name, accession, review status, expert rule and known differences.
- Declare any reviewed inserted or nested context under additional_features.
- Rebuild the reference FASTA and checksum manifest.
- Add a positive accession test and at least one related-element negative control.
- Rerun the validation ledger and complete output-bundle test.

5. Boundaries, direct repeats and validation design

How direct repeats are detected

For this group, a five-base direct repeat is supporting boundary evidence, not the naming rule. After candidate element ends have been inferred, Matryoshka reads the five bases immediately outside the left end and the five bases immediately outside the right end. An identical pair is reported with sequence and coordinates. If the record ends at the element boundary, the test is marked untestable because the flank is absent. If both flanks exist but differ, the result records that a repeat was searched for and not found.

A five-base repeat can occur by chance. It therefore cannot create a Tn1/Tn2/Tn3 name without the complete component grammar and type-profile evidence.

Validation set

Evidence class	Examples	Purpose
Reviewed exact definitions	NC_008 357, AY123 253, HM749 966, HM749 967, CP028 717, GQ160 960, V00 613	Positive controls for type/subtype execution and output generation.
Natural partial sequence	pEK499, EU935 739.1	Verify that two separated Tn2-related fragments remain partial.
Minor sequence variation	Tn1 with 25 substitutions	Verify a complete non-identical sequence becomes Tn1-like.
Structural variation	Tn2 with an 800 bp insertion	Verify an inserted sequence is retained and the call remains qualified.
Internal deletion	Tn2 with a 300 bp deletion	Verify split component evidence is chained and remains Tn2-like.
Truncation	Tn1 missing 700 left-hand bases	Verify the sequence becomes a fragment rather than exact Tn1.
Related elements	Tn1696, Tn1721, Tn5403	Verify shared components do not force a Tn1/Tn2/Tn3 name.

Headline result

All 14 real-accession ledger checks and all 23 focused Tn1/Tn2/Tn3 acceptance tests pass. The component-only tests type substituted, inserted and deleted variants without a complete-element lookup. The reviewed inputs receive exact confirmation; the pEK499 loci remain partial; related elements are not misnamed.

Validation ledger

Scenario	Record	Accession	Expected	Observed	Result
reviewed real accession	Tn1_NC_008 357	NC_008 357	Tn1	Tn1	**PASS**
reviewed real accession	Tn2_AY123 253	AY123 253	Tn2	Tn2	**PASS**
reviewed real accession	Tn3_HM749 966	HM749 966	Tn3	Tn3	**PASS**
reviewed real accession	Tn2c_HM749 967	HM749 967	Tn2c	Tn2c	**PASS**
reviewed real accession	Tn2_1_CP028 717	CP028 717	Tn2.1	Tn2.1	**PASS**
reviewed real accession	Tn1Mer_GQ160 960	GQ160 960	Tn1Mer	Tn1Mer	**PASS**
reviewed real accession	Tn3_V00 613	V00 613	Tn3 (legacy 9 bp duplication subtype)	Tn3	**PASS**
reviewed natural partial fragments	EU935 739.1	EU935 739	generic Tn1/2/3 fragment; closest Tn2	Tn1/2/3 fragment	**PASS**
reviewed natural partial fragments	EU935 739.1	EU935 739	generic Tn1/2/3 fragment; closest Tn2	Tn1/2/3 fragment	**PASS**
related but different Tn3-family element	Tn1696		no named Tn1/Tn2/Tn3 call; supported fragment permitted	no Tn1/2/3 call	**PASS**
related but different Tn3-family element	Tn1721		no named Tn1/Tn2/Tn3 call; supported fragment permitted	no Tn1/2/3 call	**PASS**
related but different Tn3-family element	Tn5403		no named Tn1/Tn2/Tn3 call; supported fragment permitted	no Tn1/2/3 call	**PASS**
synthetic minor variation	novel_complete_variant		Tn1-like, not exact Tn1	Tn1-like	**PASS**
synthetic partial sequence	left_truncated_fragment		generic Tn1/2/3 fragment; closest Tn1	Tn1/2/3 fragment	**PASS**

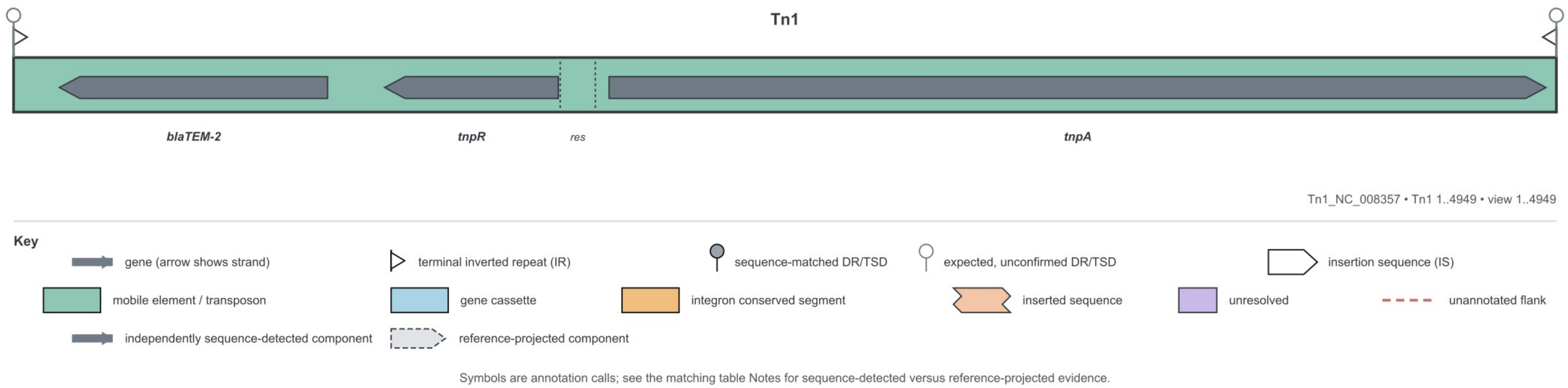
The machine-readable source is docs/validation/tn123-real-accession-results.tsv. Blank accession/type fields in the related-element rows are intentional: those tests assert that no Tn1/Tn2/Tn3 name is assigned.

6.1 Real accession result — Tn1 (NC_008357)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn1_NC_008 357	Tn1	1–4,949 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	**PASS**

Exact, complete, gap-free match to NC_008 357 plus complete component grammar.

locus map generated from the detected and assembled features for Tn1_NC_008 357.



locus table generated from the same call and evidence ledger.

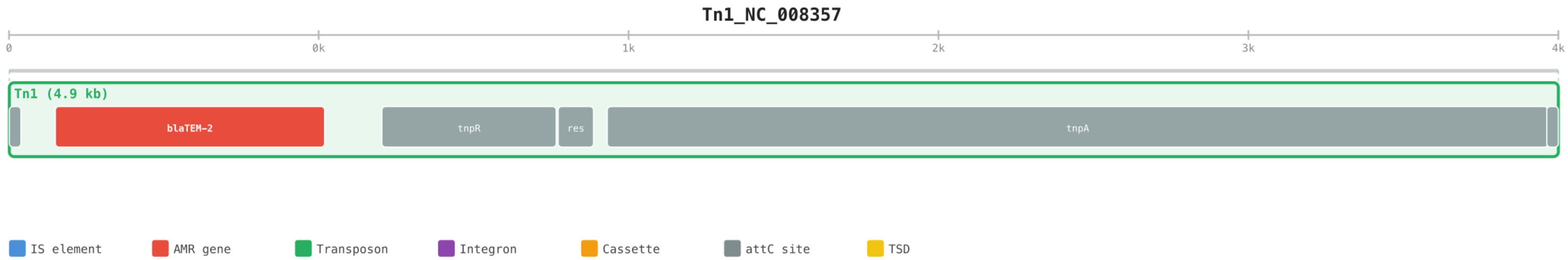
Tn1_NC_008357 • Tn1 1..4949 • view 1..4949

Position	Name*	FID	Type	Notes
1..4949	**Tn1**	NC_008357	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn1; exact reference confirmed; complete; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
148..1008	blaTEM-2		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
1191..1748	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
1754..1867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
1911..4916	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%
4912..4949	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand terminal inverted repeat (IR) sequence-matched DR/TSD expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.1 Evidence detail — Tn1 (NC_008357)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.



Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-2	148–1,008	-	100.00%	100.0%	sequence_detected
tnpR	tnpR	1,191–1,748	-	100.00%	100.0%	sequence_detected
res	res	1,754–1,867	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	1,911–4,916	+	100.00%	100.0%	sequence_detected
terminal_IR	IRR	4,912–4,949	.	100.00%	100.0%	sequence_detected

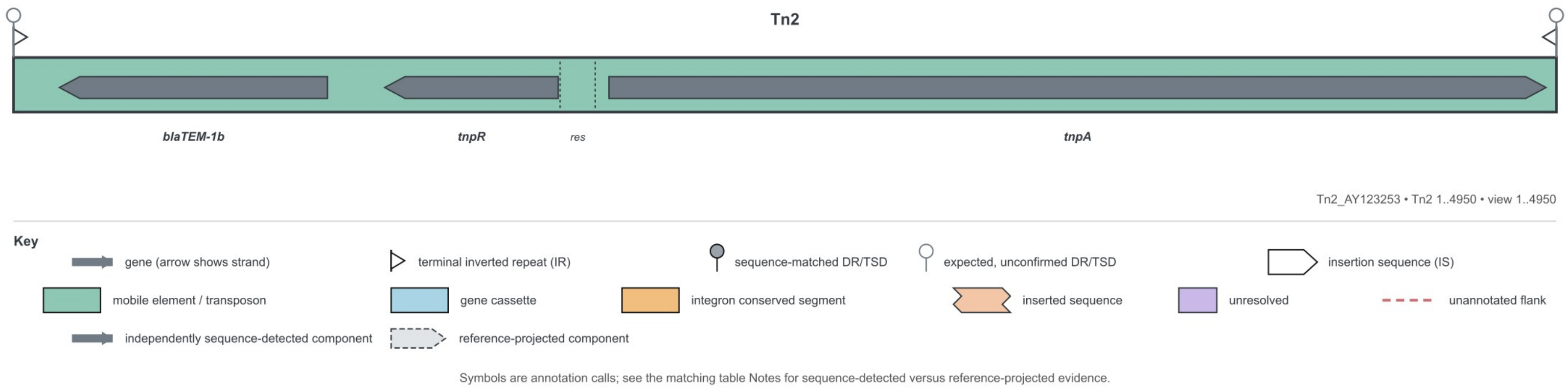
Definition: Tn1_NC_008 357 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 0.

6.2 Real accession result — Tn2 (AY123253)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn2_AY123 253	Tn2	1–4,950 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	PASS

Exact, complete, gap-free match to AY123 253 plus complete component grammar.

locus map generated from the detected and assembled features for Tn2_AY123 253.



locus table generated from the same call and evidence ledger.

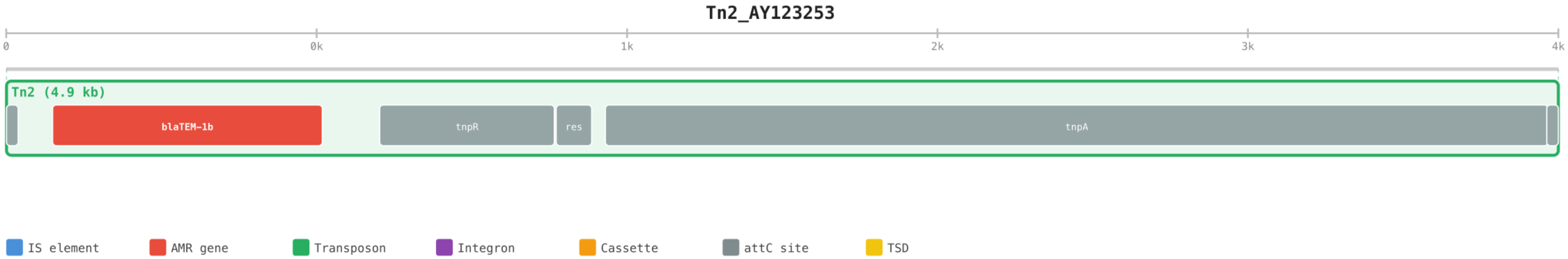
Tn2_AY123253 • Tn2 1..4950 • view 1..4950

Position	Name*	FID	Type	Notes
1..4950	**Tn2**	AY123253	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn2; exact reference confirmed; complete; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL ▷		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
148..1008	blaTEM-1b		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
1191..1748	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
1754..1867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
1911..4917	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%
4913..4950	IRR ◁		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.2 Evidence detail — Tn2 (AY123253)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.



Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-1b	148–1,008	-	100.00%	100.0%	sequence_detected
tnpR	tnpR	1,191–1,748	-	100.00%	100.0%	sequence_detected
res	res	1,754–1,867	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	1,911–4,917	+	100.00%	100.0%	sequence_detected
terminal_IR	IRR	4,913–4,950	.	100.00%	100.0%	sequence_detected

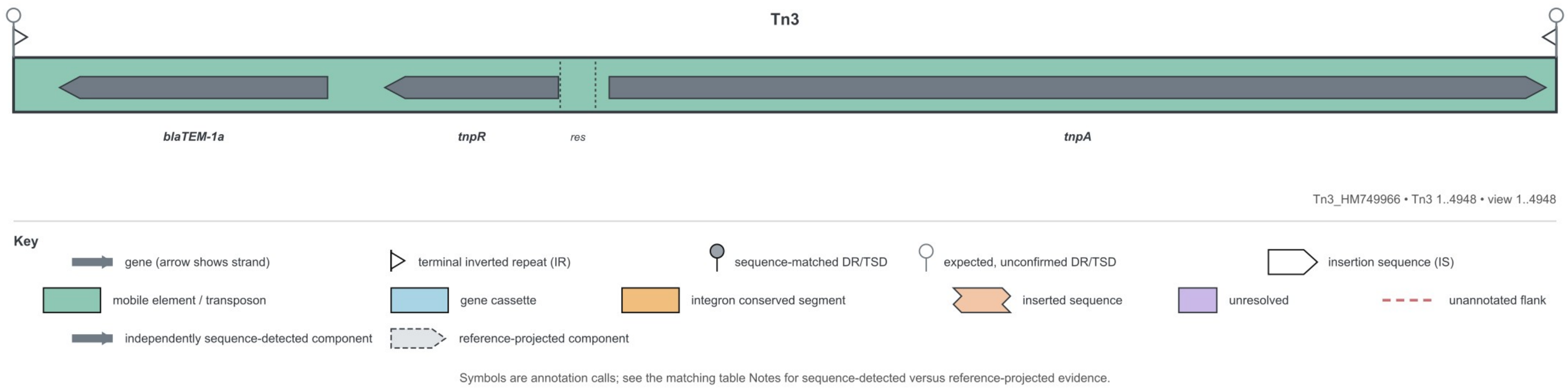
Definition: Tn2_AY123 253 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 0.

6.3 Real accession result — Tn3 (HM749966)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn3_HM749 966	Tn3	1–4,948 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	PASS

Exact, complete, gap-free match to HM749 966 plus complete component grammar.

locus map generated from the detected and assembled features for Tn3_HM749 966.



locus table generated from the same call and evidence ledger.

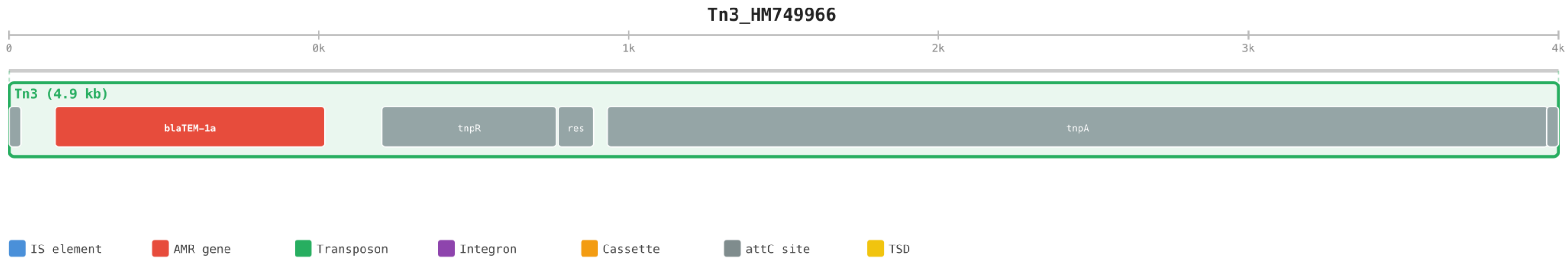
Tn3_HM749966 • Tn3 1..4948 • view 1..4948

Position	Name*	FID	Type	Notes
1..4948	**Tn3**	HM749966	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn3; exact reference confirmed; complete; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL ▷		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
148..1008	blaTEM-1a		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
1191..1748	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
1754..1867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
1911..4915	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%
4911..4948	IRR ◁		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.3 Evidence detail — Tn3 (HM749966)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.



Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-1a	148–1,008	-	100.00%	100.0%	sequence_detected
tnpR	tnpR	1,191–1,748	-	100.00%	100.0%	sequence_detected
res	res	1,754–1,867	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	1,911–4,915	+	100.00%	100.0%	sequence_detected
terminal_IR	IRR	4,911–4,948	.	100.00%	100.0%	sequence_detected

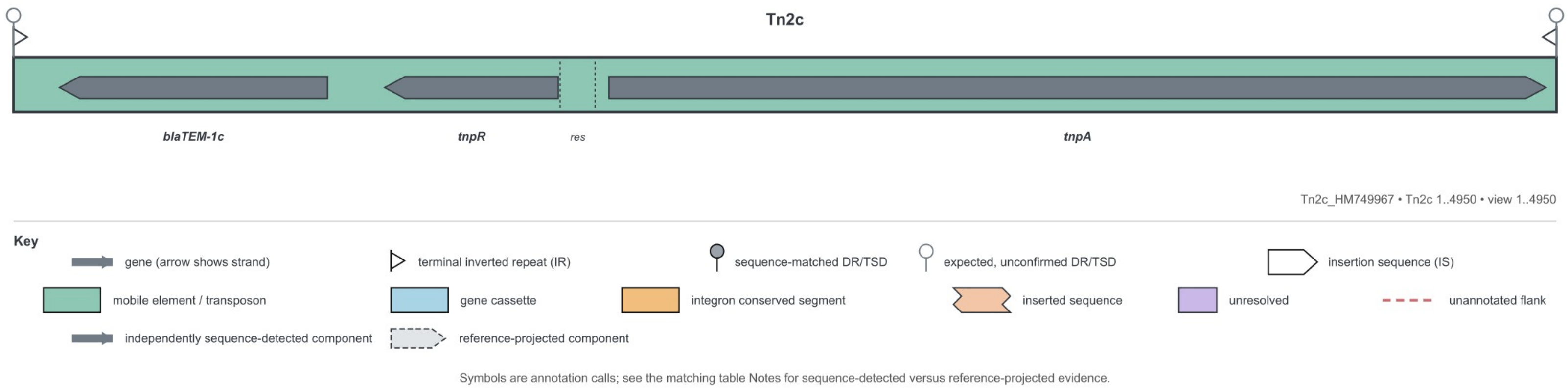
Definition: Tn3_HM749 966 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 0.

6.4 Real accession result — Tn2c (HM749967)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn2c_HM749 967	Tn2c	1–4,950 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	PASS

Exact, complete, gap-free match to the 4950 bp Tn2 interval in HM749 967. It retains the Tn2 grammar but carries the blaTEM-1c sequence and nine substitutions relative to AY123 253.

locus map generated from the detected and assembled features for Tn2c_HM749 967.



locus table generated from the same call and evidence ledger.

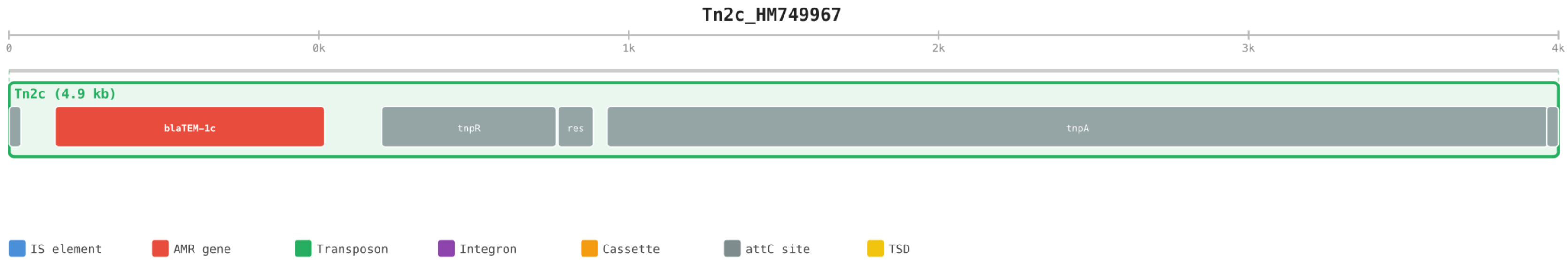
Tn2c_HM749967 • Tn2c 1..4950 • view 1..4950

Position	Name*	FID	Type	Notes
1..4950	**Tn2c**	HM749967	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn2c (subtype Tn2c); exact reference confirmed; complete; reviewed subtype differences: 9 nucleotide substitutions relative to canonical Tn2 AY123253, blaTEM-1c allele; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL ▷		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
148..1008	blaTEM-1c		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
1191..1748	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
1754..1867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
1911..4917	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%
4913..4950	IRR ◁		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.4 Evidence detail — Tn2c (HM749967)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.



Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-1c	148–1,008	-	100.00%	100.0%	sequence_detected
tnpR	tnpR	1,191–1,748	-	100.00%	100.0%	sequence_detected
res	res	1,754–1,867	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	1,911–4,917	+	100.00%	100.0%	sequence_detected
terminal_IR	IRR	4,913–4,950	.	100.00%	100.0%	sequence_detected

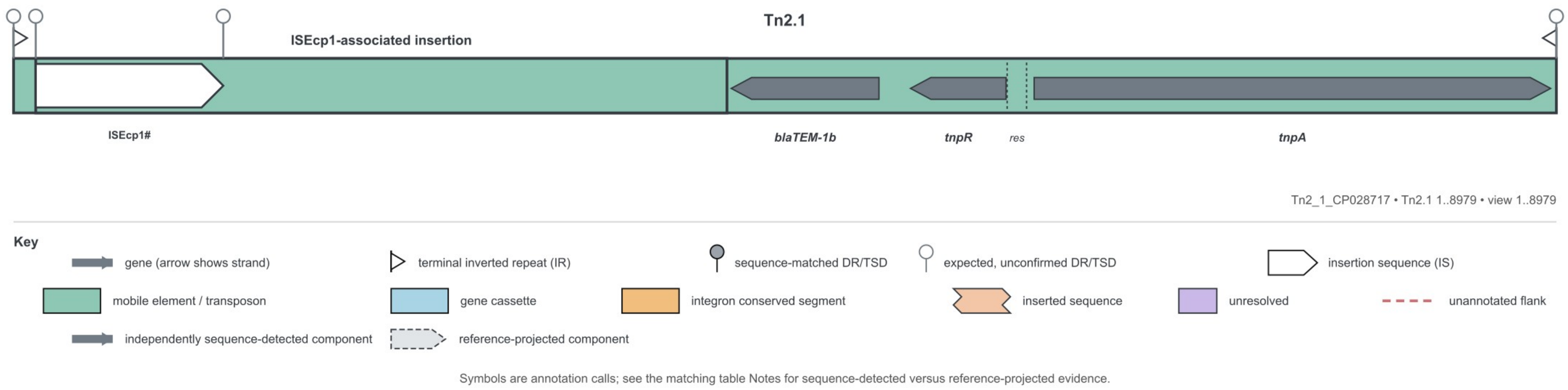
Definition: Tn2c_HM749 967 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 0.

6.5 Real accession result — Tn2.1 (CP028717)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn2_1_CP028 717	Tn2.1	1–8,979 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	PASS

Exact match to the 8979 bp interrupted Tn2 structure extracted from CP028 717. The Tn2 backbone is complete, including both terminal IRs, and a 4024 bp ISEcp1-associated region interrupts the sequence between canonical reference positions 124 and 129.

locus map generated from the detected and assembled features for Tn2_1_CP028 717.



6.5 Continued — Tn2.1 locus table

locus table generated from the same call and evidence ledger.

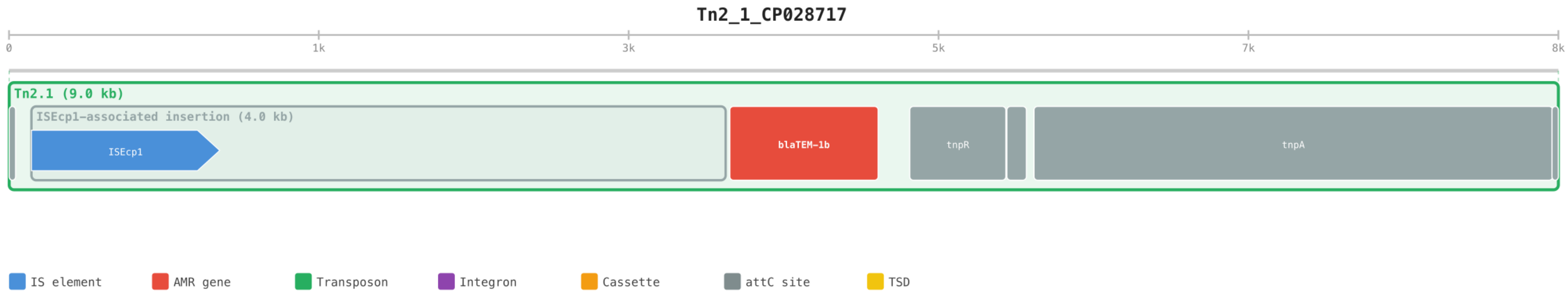
Tn2_1_CP028717 • Tn2.1 1..8979 • view 1..8979

Position	Name*	FID	Type	Notes
1..8979	**Tn2.1**	CP028717	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn2.1 (subtype Tn2.1); exact reference confirmed; complete; reviewed subtype differences: 4024 bp ISEcp1-associated insertion near the blaTEM end, 5 bp duplicated Tn2 sequence flanks the insertion junction; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
130..4153	**ISEcp1-associated insertion**	CP028717	TPU	
130..1221	ISEcp1	FJ621588	IS	
4177..5037	blaTEM-1b		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
5220..5777	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
5783..5896	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
5940..8946	tnpA		gene	independently sequence-detected tnpA; identity=99.97%; coverage=100.0%
8942..8979	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.5 Evidence detail — Tn2.1 (CP028717)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.

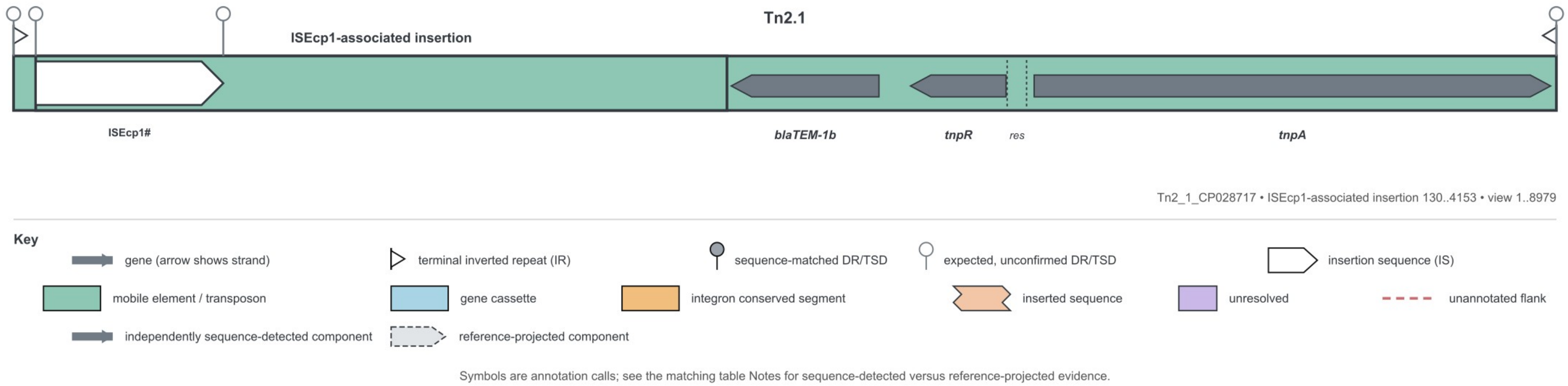


Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-1b	4,177–5,037	-	100.00%	100.0%	sequence_detected
tnpR	tnpR	5,220–5,777	-	100.00%	100.0%	sequence_detected
res	res	5,783–5,896	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	5,940–8,946	+	99.97%	100.0%	sequence_detected
terminal_IR	IRR	8,942–8,979	.	100.00%	100.0%	sequence_detected

Definition: Tn2_1_CP028 717 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 2.

6.5 Nested locus — ISEcp1-associated insertion

The interrupted Tn2.1 definition contains a separately indexed ISEcp1-associated locus. It is drawn independently as well as inside the whole Tn2.1 view.



Tn2_1_CP028717 • ISEcp1-associated insertion 130..4153 • view 1..8979

Position	Name*	FID	Type	Notes
1..8979	**Tn2.1**	CP028717	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn2.1 (subtype Tn2.1); exact reference confirmed; complete; reviewed subtype differences: 4024 bp ISEcp1-associated insertion near the blaTEM end, 5 bp duplicated Tn2 sequence flanks the insertion junction; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
130..4153	**ISEcp1-associated insertion**	CP028717	TPU	
130..1221	ISEcp1	FJ621588	IS	
4177..5037	blaTEM-1b		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
5220..5777	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
5783..5896	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
5940..8946	tnpA		gene	independently sequence-detected tnpA; identity=99.97%; coverage=100.0%
8942..8979	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

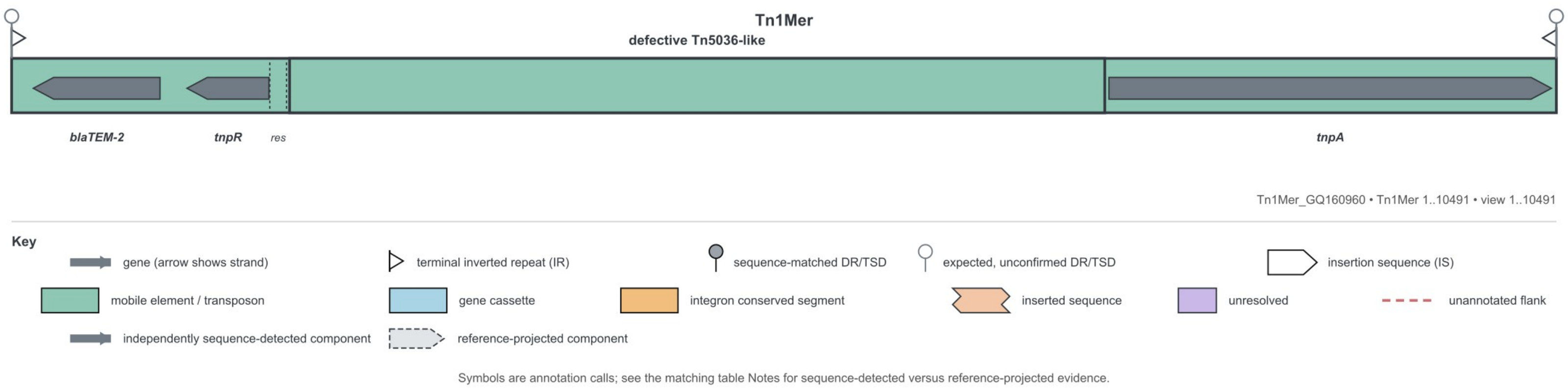
Key: strand terminal inverted repeat (IR) sequence-matched DR/TSD expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.6 Real accession result — Tn1Mer (GQ160960)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn1Mer_GQ160 960	Tn1Mer	1–10,491 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	**PASS**

Exact match to the 10 491 bp interrupted Tn1 structure in GQ160 960. The outer Tn1 grammar is complete, while a defective 5537 bp Tn5036-like transposon interrupts the Tn1 backbone. This is a named interrupted subtype, not the canonical Tn1 reference.

locus map generated from the detected and assembled features for Tn1Mer_GQ160 960.



locus table generated from the same call and evidence ledger.

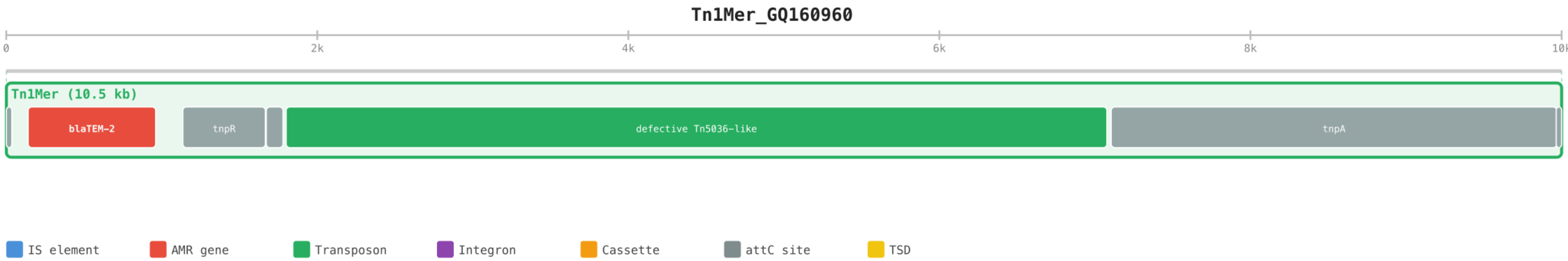
Tn1Mer_GQ160960 • Tn1Mer 1..10491 • view 1..10491

Position	Name*	FID	Type	Notes
1..10491	**Tn1Mer**	GQ160960	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn1Mer (subtype Tn1Mer); exact reference confirmed; complete; reviewed subtype differences: defective Tn5036-like transposon inserted into the Tn1 backbone, blaTEM-1 rather than canonical Tn1 blaTEM-2; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
148..1008	blaTEM-2		R_gene	independently sequence-detected blaTEM; identity=99.65%; coverage=100.0%
1191..1748	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
1754..1867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
1888..7424	**defective Tn5036-like**	GQ160960	Tn	interrupted
7453..10458	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%
10454..10491	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand; terminal inverted repeat (IR); sequence-matched DR/TSD; expected, unconfirmed DR/TSD. Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.6 Evidence detail — Tn1Mer (GQ160960)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.



Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-2	148–1,008	-	99.65%	100.0%	sequence_detected
tnpR	tnpR	1,191–1,748	-	100.00%	100.0%	sequence_detected
res	res	1,754–1,867	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	7,453–10,458	+	100.00%	100.0%	sequence_detected
terminal_IR	IRR	10,454–10,491	.	100.00%	100.0%	sequence_detected

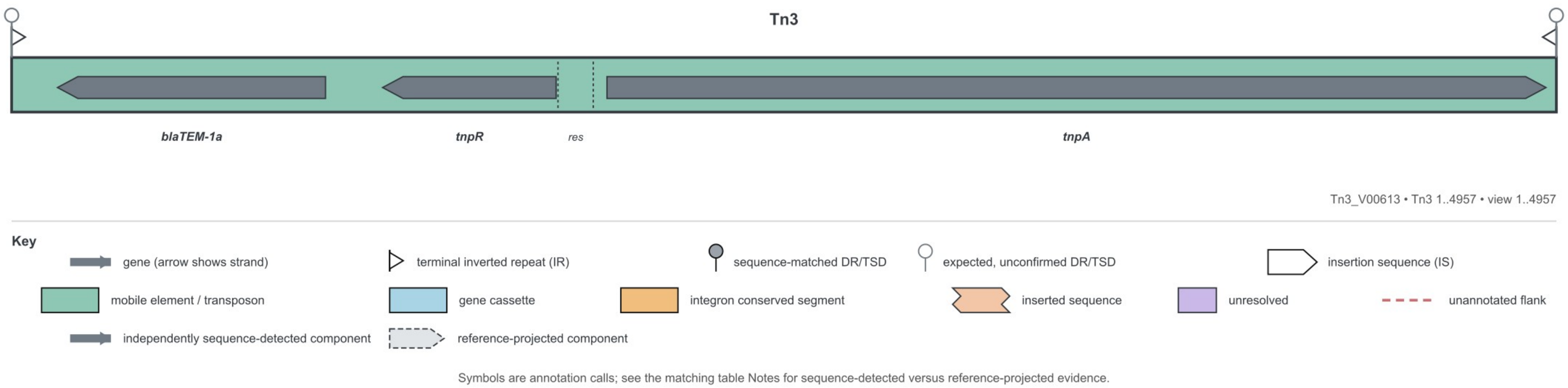
Definition: Tn1Mer_GQ160 960 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 1.

6.7 Real accession result — Tn3 (V00613)

Input record	Observed call	Coordinates	Whole-locus match	Grammar	Proof
Tn3_V00 613	Tn3	1–4,957 (+)	100.0% identity; 100.0% coverage	6/6 detected; order valid	PASS

Exact match to V00 613 is Tn3 with the declared 9 bp duplication. It is a recognised Tn3 sequence variant, not an exact match to canonical HM749 966.

locus map generated from the detected and assembled features for Tn3_V00 613.



locus table generated from the same call and evidence ledger.

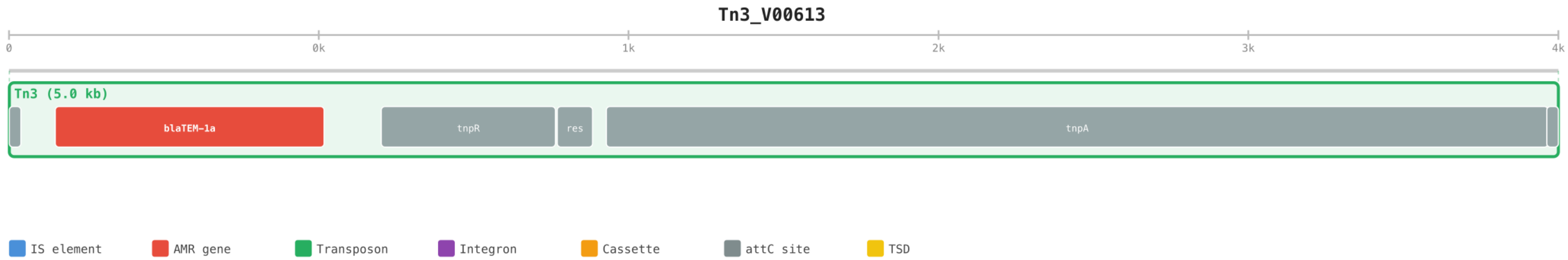
Tn3_V00613 • Tn3 1..4957 • view 1..4957

Position	Name*	FID	Type	Notes
1..4957	Tn3	V00613	Tn	BLAST identity=100%; reference coverage=100%; closest definition=Tn3 (subtype V00613_legacy_9bp_duplication); exact reference confirmed; complete with insertion; inserted sequence≈9 bp; reviewed subtype differences: 9 bp duplication relative to canonical Tn3 HM749966; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; flanking sequence unavailable
1..38	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
148..1008	blaTEM-1a		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
1191..1748	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
1754..1867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
1911..4924	tnpA		gene	independently sequence-detected tnpA; identity=99.7%; coverage=100.0%; complete with insertion
4920..4957	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand — terminal inverted repeat (IR) — sequence-matched DR/TSD — expected, unconfirmed DR/TSD — Solid components are sequence-detected; dashed/outlined components are reference-projected.

6.7 Evidence detail — Tn3 (V00613)

Original scale-accurate hierarchical view. Parent-child nesting is preserved.



Role	Feature	Coordinates	Strand	Identity	Coverage	Evidence
terminal_IR	IRL	1–38	.	100.00%	100.0%	sequence_detected
blaTEM	blaTEM-1a	148–1,008	-	100.00%	100.0%	sequence_detected
tnpR	tnpR	1,191–1,748	-	100.00%	100.0%	sequence_detected
res	res	1,754–1,867	.	100.00%	100.0%	sequence_detected
tnpA	tnpA	1,911–4,924	+	99.70%	100.0%	sequence_detected
terminal_IR	IRR	4,920–4,957	.	100.00%	100.0%	sequence_detected

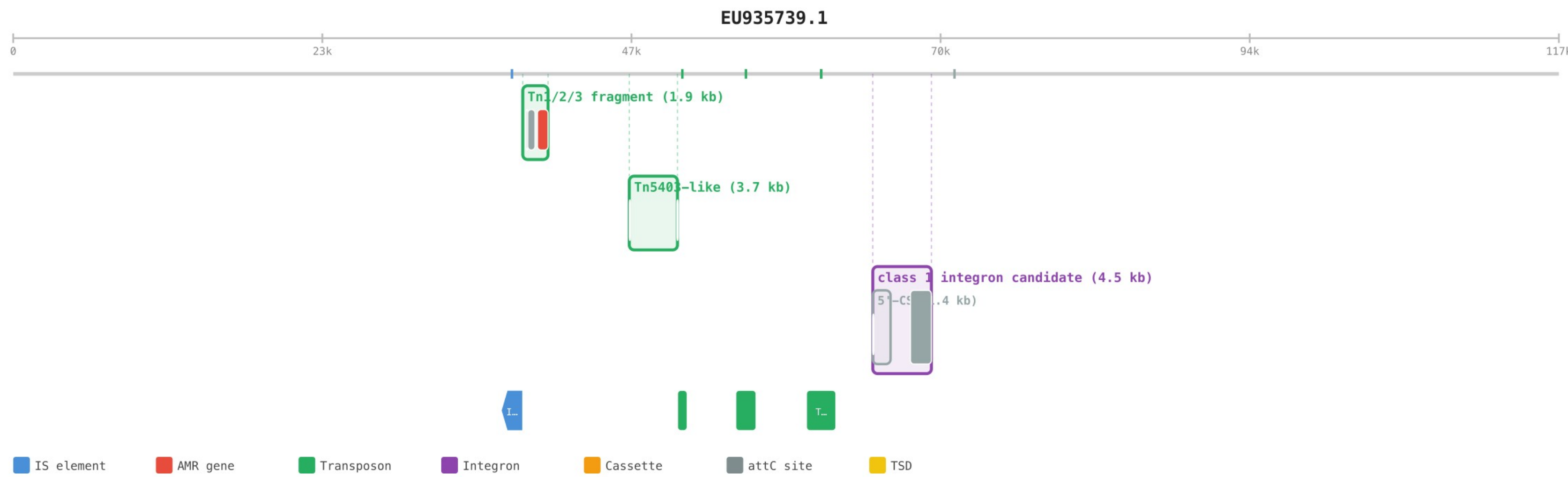
Definition: Tn3_V00 613 (version 2026-08-27). Variant status: exact_reference_confirmed. Reference-projected required components: 0. Context features from the reviewed definition: 0.

7. Natural partial-fragment result — pEK499 (EU935739.1)

Coordinates	Observed call	Closest type	Reference coverage	Detected required roles	Proof
38,747–40,671	Tn1/2/3 fragment	Tn2	38.89%	blaTEM, tnpR, res	PARTIAL
60,316–62,561	Tn1/2/3 fragment	Tn2	45.37%	tnpA	PARTIAL

This is the key natural-context behaviour: two separated Tn2-related regions are retained independently. Neither contains both terminal ends and all required components, so the software does not join them or call an exact Tn2.

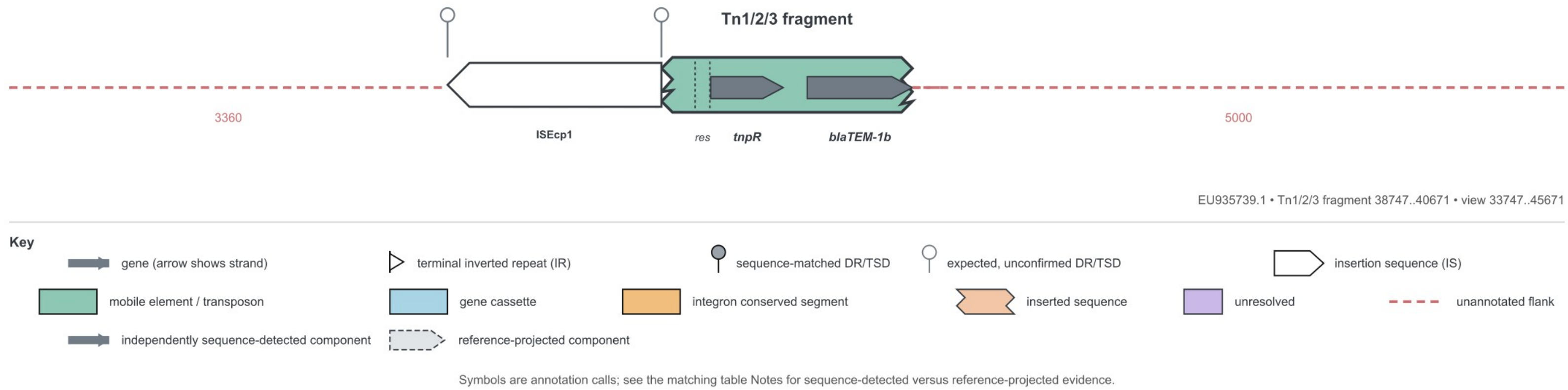
Original hierarchy view for the complete 117,536 bp pEK499 record.



7.1 pEK499 fragment 38,747–40,671

Observed call: Tn1/2/3 fragment; closest reviewed type: Tn2; identity 100.0%; reference coverage 38.89%; proof verdict PARTIAL.

locus map. Missing element ends/components remain visibly absent.



locus table for the partial call. The partial status and incomplete grammar are explicit.

EU935739.1 • Tn1/2/3 fragment 38747..40671 • view 33747..45671

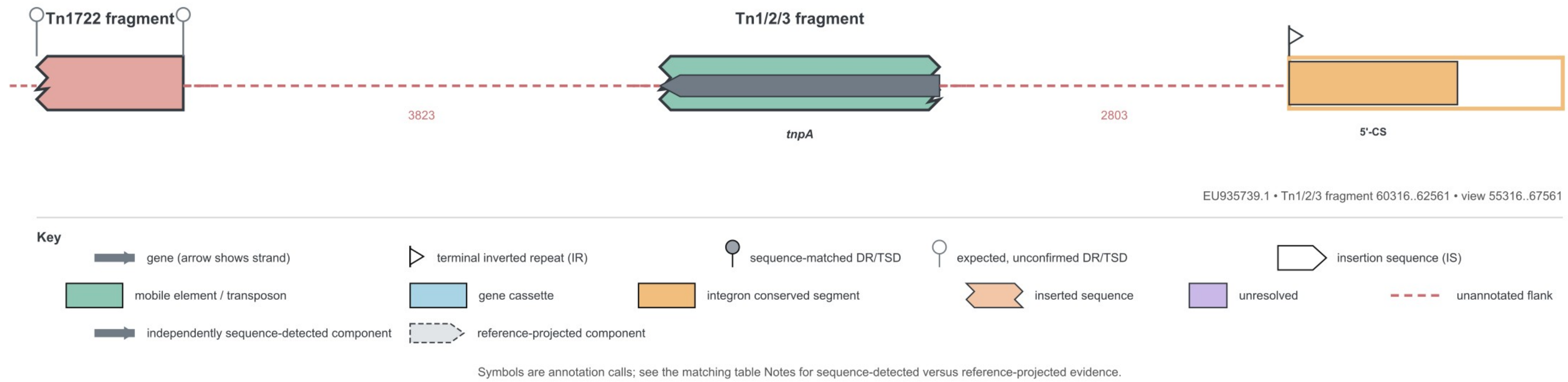
Position	Name*	FID	Type	Notes
37107..38746	ISEcp1	FJ621588	IS	
38747..40671	**Tn1/2/3 fragment**	AY123253	Tn	BLAST identity=100%; reference coverage=38.89%; closest definition=Tn2; rule incomplete; partial or conflicting; component grammar=partial_or_conflicting; 3 independently sequence-detected components; exact Tn1/Tn2/Tn3 identity unresolved
39005..39118	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
39124..39681	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
39864..40671	blaTEM-1b		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=93.84%; left partial

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

7.2 pEK499 fragment 60,316–62,561

Observed call: Tn1/2/3 fragment; closest reviewed type: Tn2; identity 100.0%; reference coverage 45.37%; proof verdict PARTIAL.

locus map. Missing element ends/components remain visibly absent.



locus table for the partial call. The partial status and incomplete grammar are explicit.

EU935739.1 • Tn1/2/3 fragment 60316..62561 • view 55316..67561

Position	Name*	FID	Type	Notes
54940..56492	**Tn1722 fragment**	LOCUS-REF-Tn1722_AB366441	Tn	BLAST identity=99.94%; reference coverage=27.54%; left partial; left partial; expected DR=5 bp; searched but not found; exact Tn1/Tn2/Tn3 identity unresolved
60316..62561	**Tn1/2/3 fragment**	AY123253	Tn	BLAST identity=100%; reference coverage=45.37%; closest definition=Tn2; rule incomplete; partial or conflicting; component grammar=partial_or_conflicting; 1 independently sequence-detected components; exact Tn1/Tn2/Tn3 identity unresolved
60316..62561	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=74.69%; internal fragment
65365..69821	class 1 integron candidate		region	assembled from exact locus component references
65365..66716	5'-CS	LOCUS-REF-5CS-standard	CS	
65365..65389	IRi	LOCUS-REF-IRi	IR	terminal inverted repeat; IR=25 bp

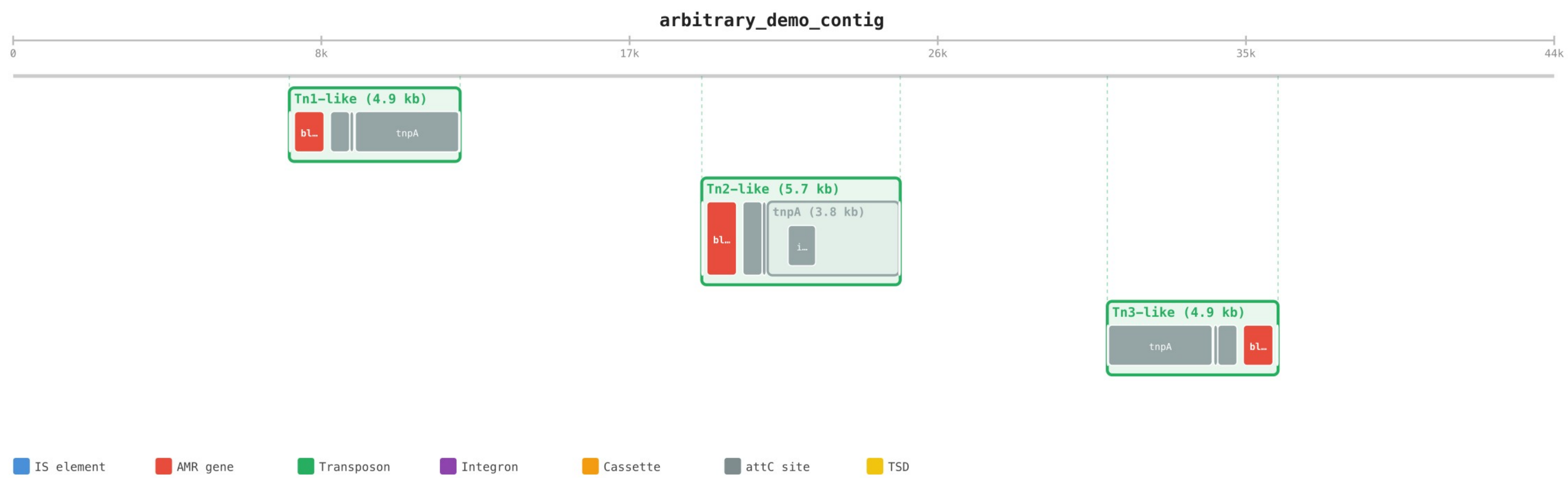
Key: strand ▷ terminal inverted repeat (IR) sequence-matched DR/TSD expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

8. Arbitrary-sequence demonstration

A single 44,647 bp contig was constructed with unrelated flanking sequence and three loci: a Tn1-derived sequence with 25 substitutions, a Tn2-derived sequence with an approximately 800 bp internal insertion, and a reverse-complement Tn3-derived sequence. It was processed with --profile tn123-components, which excludes complete Tn1/Tn2/Tn3 reference lookup. Every biological call below therefore comes from detected components and expert rules.

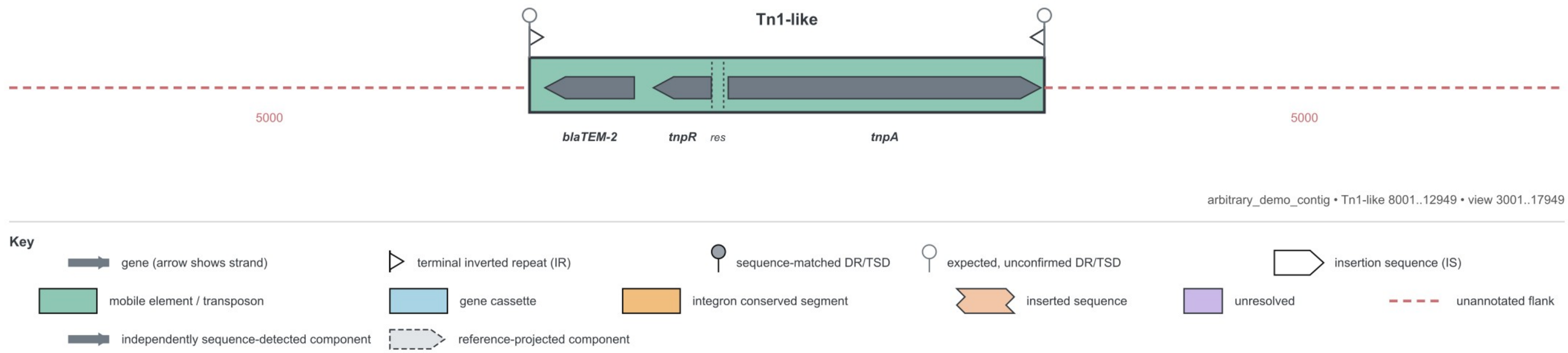
Coordinates	Call	Strand	Rule type	Component score	Whole-element lookup	Proof
8,001–12,949	Tn1-like	+	Tn1	99.592	none	**PASS**
19,950–25,699	Tn2-like	+	Tn2	100.0	none	**PASS**
31,700–36,647	Tn3-like	-	Tn3	100.0	none	**PASS**

Original hierarchy view: all three loci are retained on their shared contig and nested features remain visible.



8.1 Arbitrary contig locus — Tn1-like

The component grammar is complete and the expert component-profile rules assign Tn1. No complete Tn1/Tn2/Tn3 reference comparison was run; the visible call is Tn1-like. Insertions and other structural differences are retained from split component alignments and shown in the map/table.



Symbols are annotation calls; see the matching table Notes for sequence-detected versus reference-projected evidence.

arbitrary_demo_contig • Tn1-like 8001..12949 • view 3001..17949

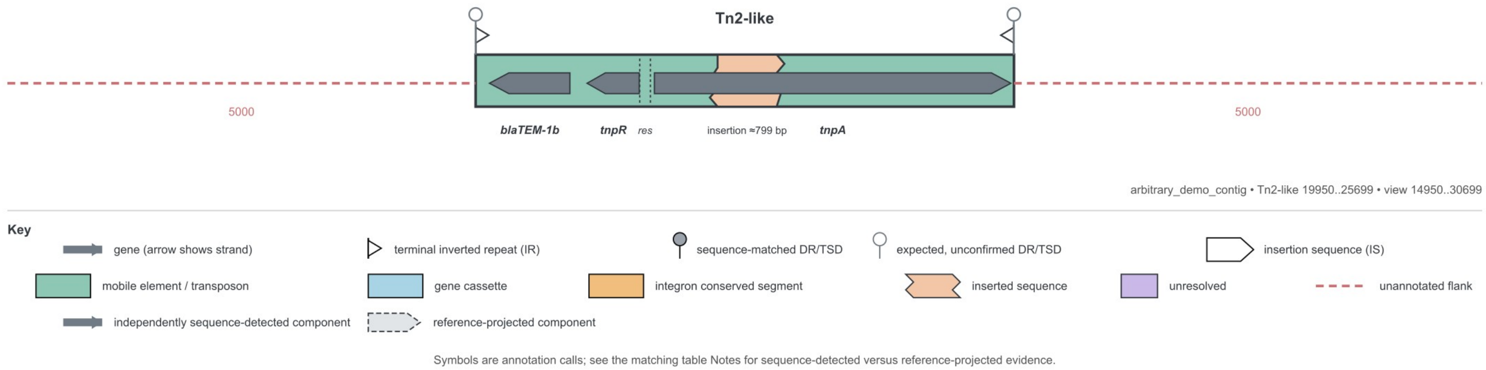
Position	Name*	FID	Type	Notes
8001..12949	**Tn1-like**		Tn	rule based type candidate; complete; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; searched but not found
8001..8038	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
8148..9008	blaTEM-2		R_gene	independently sequence-detected blaTEM; identity=99.42%; coverage=100.0%
9191..9748	tnpR		gene	independently sequence-detected tnpR; identity=99.1%; coverage=100.0%
9754..9867	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
9911..12916	tnpA		gene	independently sequence-detected tnpA; identity=99.57%; coverage=100.0%
12912..12949	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD

Solid components are sequence-detected; dashed/outlined components are reference-projected.

8.2 Arbitrary contig locus — Tn2-like

The component grammar is complete and the expert component-profile rules assign Tn2. No complete Tn1/Tn2/Tn3 reference comparison was run; the visible call is Tn2-like. Insertions and other structural differences are retained from split component alignments and shown in the map/table.



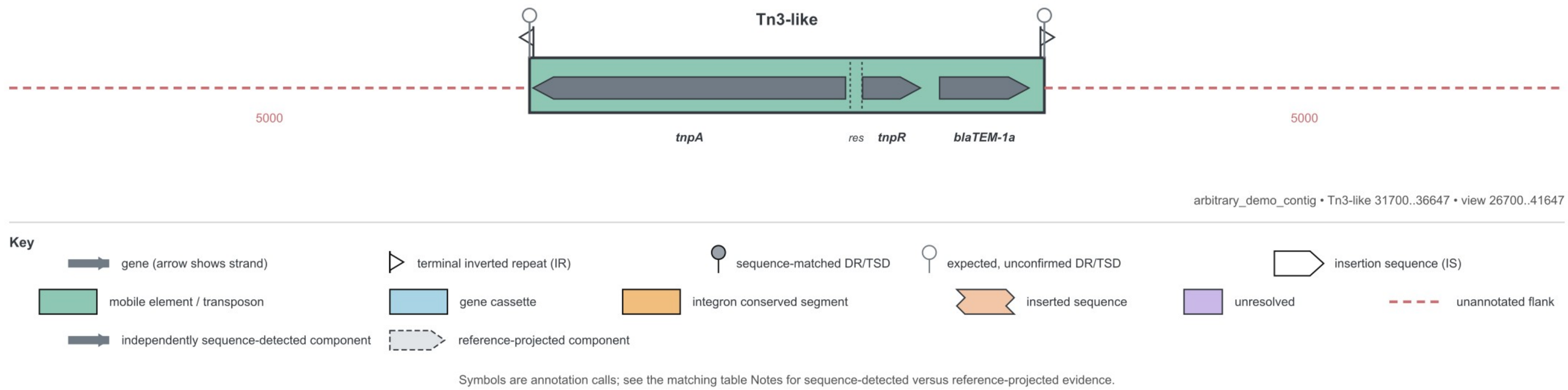
arbitrary_demo_contig • Tn2-like 19950..25699 • view 14950..30699

Position	Name*	FID	Type	Notes
19950..25699	**Tn2-like**		Tn	rule based type candidate; complete with insertion; inserted sequence≈799 bp; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; searched but not found
19950..19987	IRL ▷		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
20097..20957	blaTEM-1b		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
21140..21697	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
21703..21816	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
21860..25666	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%; complete with insertion
22451..23249	inserted sequence 1		inserted	unclassified insertion; estimated inserted sequence=799 bp
25662..25699	IRR ◁		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand ▷ terminal inverted repeat (IR) ● sequence-matched DR/TSD ○ expected, unconfirmed DR/TSD Solid components are sequence-detected; dashed/outlined components are reference-projected.

8.3 Arbitrary contig locus — Tn3-like

The component grammar is complete and the expert component-profile rules assign Tn3. No complete Tn1/Tn2/Tn3 reference comparison was run; the visible call is Tn3-like. Insertions and other structural differences are retained from split component alignments and shown in the map/table.



arbitrary_demo_contig • Tn3-like 31700..36647 • view 26700..41647

Position	Name*	FID	Type	Notes
31700..36647	**Tn3-like**		Tn	rule based type candidate; complete; component grammar=complete; 6 independently sequence-detected components; expected DR=5 bp; searched but not found
31700..31737	IRR		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%
31733..34737	tnpA		gene	independently sequence-detected tnpA; identity=100.0%; coverage=100.0%
34781..34894	res		res	independently sequence-detected res; identity=100.0%; coverage=100.0%
34900..35457	tnpR		gene	independently sequence-detected tnpR; identity=100.0%; coverage=100.0%
35640..36500	blaTEM-1a		R_gene	independently sequence-detected blaTEM; identity=100.0%; coverage=100.0%
36610..36647	IRL		IR	independently sequence-detected terminal_IR; identity=100.0%; coverage=100.0%

Key: strand   terminal inverted repeat (IR)   sequence-matched DR/TSD   expected, unconfirmed DR/TSD   Solid components are sequence-detected; dashed/outlined components are reference-projected.

9. Output package and interpretation

Output	Purpose
annotation.json	Complete machine-readable hierarchy, identifiers, coordinates, evidence, definitions and confidence.
annotation.gff3	Interoperable genomic feature annotation.
annotation.cell	Original nested CellGen/Wolvercote representation.
hierarchy/*.svg	Original scale-accurate parent-child view of the record.
locus-map/*.svg	Readable locus-centred locus map with its own legend/key.
locus-table/*.svg	locus feature/evidence table for the same locus.
proof/report.html	Linked human-readable evidence report.
proof/proof.json; components.tsv; matches.tsv	Machine-checkable component-to-call ledger.
run.json	Run metadata plus direct paths to every generated locus map and table.

CellGen example

```
({{}IRL, {}blaTEM-2, {}tnpR, {}res, {}tnpA, {}IRR}Tn1[family="Tn1"])Tn1_NC_008 357
```

The CellGen line carries the same nesting as the hierarchy graphic: the six components are children of Tn1, which belongs to the input record.

How to read the figures

- The locus maps prioritises biological readability around one locus.
- The locus table is the feature-by-feature description, including evidence and call qualification.
- The hierarchy view preserves scale and parent-child nesting across the whole record.
- The proof ledger separates sequence-detected required components from context added from a reviewed subtype definition.
- The legend/key is embedded in every locus map and table so the symbols are interpretable without prior experience of the notation.

10. Limitations and what should happen next

What is proven now

The implementation can recover the defined components, apply reviewable Tn1/Tn2/Tn3 rules, distinguish exact from qualified and partial calls, and generate the complete set of requested outputs on the included controls and demonstrations.

What is not yet proven

- Population-level sensitivity and specificity across a large independent accession panel.
- Exhaustive subtype coverage or complete parity with every component and compound element described across the source literature.
- Assigning a new formal element name to a previously undescribed structure without expert review; novel candidates within the defined family are deliberately reported as type-like.

- Target-site duplication evidence when the input sequence ends exactly at the element boundary or lacks sufficient flank.
- Consistent optional-detector availability on every installation; the built-in validated scan is the portable core.

Recommended next validation

Have a domain expert review the human-readable definitions and embedded figures; freeze accepted naming decisions; then evaluate a blinded accession set containing true Tn1/Tn2/Tn3 examples, close variants, fragmented assemblies and related Tn3-family elements. Report locus-level precision/recall, boundary error and exact-versus-qualified naming accuracy.

Expert decisions still requested

- Confirm whether HM749 967 should be displayed as Tn2c.
- Confirm the preferred name and internal labels for the CP028 717 interrupted Tn2.1 structure.
- Confirm whether Tn1Mer is a public named subtype or an interrupted Tn1 example.
- Confirm the preferred wording for V00 613 relative to canonical HM749 966.
- Supply or approve additional type/subtype definitions and negative controls for the next expansion.